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RESEARCH ARTICLE

Sleep Apnea Detection from Respiratory Sounds Using Time–Frequency Features and Neural Networks

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Abstract

Sleep apnea is a prevalent and frequently underdiagnosed disorder whose diagnosis depends on polysomnography, a costly and limited resource that hinders large-scale screening. Although recent AI-based approaches have explored signals such as ECG, PPG, or EEG, most remain focused on obstructive apnea, rely on complex architectures, or lack validation using accessible data suitable for home monitoring. This study proposes a lightweight deep learning model that detects apnea events from respiratory audio transformed into time–frequency features. Using MFCC, Mel-spectrogram, and energy-based descriptors, the proposed model achieved an accuracy of 85.84%, an F1-score of 88.96%, and an AUC of 93.06% on the independent test set, while requiring only a single acoustic channel. The results demonstrate that respiratory sounds contain sufficient information for reliable apnea detection, supporting their use as a non-invasive, low-cost alternative for scalable home-based screening and telemonitoring applications.

KEYWORDS

Sleep apnea, deep learning, respiratory sounds, feature extraction, neural networks

1 | INTRODUCTION

Sleep apnea comprises two main types: obstructive sleep apnea (OSA) and central sleep apnea (CSA). This disorder is characterized by repeated episodes of breathing interruption during the sleep cycle, leading to significant sleep fragmentation. Its cardiovascular, metabolic, and neurocognitive consequences are widely recognized (Gosselin et al. 2019, Quironsalud 2025). Although polysomnography (PSG) remains the gold standard for diagnosis, its high cost, complexity, and limited availability have driven the search for automated and low-cost solutions based on minimal signals and deep learning algorithms for home-based diagnosis and monitoring (Zhang et al. 2025, Attia et al. 2025).

Recent research has shown an increasing use of artificial intelligence methods, particularly convolutional neural networks (CNN) and hybrid architectures applied to signals such as electrocardiogram (ECG), photoplethysmography (PPG), nasal pressure, thoracoabdominal effort, and single-channel electroencephalogram (EEG). These approaches have demonstrated the clinical feasibility of using reduced signals for the detection and estimation of the apnea–hypopnea index (AHI). Representative examples include OSA detection and assessment from ECG using deep learning (Zhang et al. 2025, Kumar et al. 2025), clinical validations of models based on PPG signals in wearable devices (Attia et al. 2025), and OSA estimation through heart rate variability (HRV) using CNNs (Padovano et al. 2025). In addition, multimodal models combining EEG, ECG, and respiratory flow through time–frequency transforms and CNN-based fusion have been explored to distinguish respiratory event subtypes (Ghafourian et al. 2025, Varshini et al. 2024).

In parallel, studies focusing on thoracic and abdominal effort signals have demonstrated their potential to characterize central events, enabling the estimation of the central apnea–hypopnea index using 2D-CNN models and emphasizing the importance of measuring respiratory effort to differentiate CSA and OSA (Barroso-García et al. 2023). Likewise, research in pediatric populations has shown the feasibility of predicting imminent apnea events, for instance, from nasal pressure using CNN and transfer learning, as well as estimating severity from pulse oximetry (SpO_2) using ResNet or CNN-RNN architectures. These findings open the possibility of preventive strategies based on real-time anticipation of respiratory events (Crowson et al. 2023, Mortazavi et al. 2024, Chaw et al. 2023).

Regarding methodology, three predominant trends have been identified: (i) the use of hybrid architectures (CNNs for spatiotemporal feature extraction combined with sequential blocks such as BiLSTM, BiGRU, or

Abbreviations:

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Transformers with attention mechanisms) to capture temporal dependencies and sleep-stage context (Mortazavi et al. 2024, Saha et al. 2024, Nandakumar et al. 2025); (ii) the use of bidimensional representations (spectrograms or recurrence plots) and multimodal fusion for robustness against noise and sensor variability (Padovano et al. 2025, Nandakumar et al. 2025, Javeed et al. 2023); and (iii) a growing interest in external validation and real-time clinical studies, through developments validated in real environments or non-intrusive devices such as PPG-based or pulse oximetry platforms and *smart-bed* systems, allowing models trained on large databases to be deployed in the field, although most remain focused on OSA rather than CSA (Attia et al. 2025, Siyahjani et al. 2025).

Despite these advances, important gaps persist that hinder progress in the early detection and prevention of CSA. Among them are the scarcity of public datasets with detailed annotations, the frequent use of 30-second segments that may smooth pre-event signals, and the lack of validation in adults with clinical interpretability. Few studies validate predictive CSA models in adults, which limits their practical applicability (Padovano et al. 2025, Barroso-García et al. 2023, Javeed et al. 2023). Moreover, research based on electronic clinical data and hybrid models such as XGBoost–BiLSTM has shown potential to predict population-level risk using demographic and clinical history features, suggesting a valuable complement to signal-based methods (Javeed et al. 2023).

Recently, various studies have sought non-invasive and cost-effective methods for diagnosing this disorder using acoustic signals, mainly tracheal respiratory sounds or snoring (Res 2025). It has been observed that acoustic patterns show clear differences between obstructive (OSA) and, to a lesser extent, central (CSA) apnea. However, polysomnography (PSG) remains the gold standard, but the search for more portable and accessible alternatives has motivated the study of sounds generated by snoring, which originate in the upper airways (e.g., tongue, palate, and pharyngeal wall). These characteristics suggest that such sounds contain essential information about airway patency (Wang et al. 2022). Likewise, the nature of respiratory sounds during sleep varies according to the degree of airway obstruction, although their acquisition may be affected by environmental noise or recording interference. Despite these limitations, audio-based methods hold great potential, as they could be integrated into common devices such as mobile phones or smart speakers to provide continuous and contactless monitoring of patients (Le et al. 2023).

In recent works, some studies have achieved significant progress through deep learning models applied to nasal pressure airflow signals. A representative example is the OSASS model, which employs a multi-resolution residual network (Mr-ResNet) architecture to process single-channel nasal pressure signals. This approach achieved an overall accuracy above 91%, with sensitivities and specificities around 90%, comparable to those obtained via polysomnography, for detecting the severity of obstructive sleep apnea (OSA) (Yue et al. 2021). However, such approaches are typically limited to OSA detection.

On the other hand, other methodologies have explored predictive models based on clinical and demographic data, focusing particularly on diagnostic methods and their optimization. Current tools for OSA detection often face challenges such as low examination efficiency and high cost. Consequently, machine learning approaches capable of continuous learning and iterative improvement have been developed. These models enable the identification of obstructive apnea (OSA) in early or intermediate stages, facilitating timely clinical interventions (Liu et al. 2024). Although such approaches do not detect respiratory events during sleep, they offer a useful complement for prioritizing patients who may require more complex diagnostic evaluations.

More recent studies have explored multimodal approaches that combine multiple physiological signals such as ECG, PPG, and respiratory flow. This combination improves accuracy and robustness against noise or data loss. For example, it has been shown that combining electrocardiographic and respiratory signals through CNN–Transformer architectures yields more stable detection for both obstructive and central apnea, reducing false positives compared to single-signal models. These methodologies aim to leverage the complementary information of each source, thereby improving the classification of different respiratory events during sleep (Li et al. 2023).

In particular, recent studies have explored portable monitors (PMs) that integrate deep learning models for OSA detection. For example, a 1D-CNN trained on PSG-derived data for low-cost devices achieved 84.3% accuracy and used Grad-CAM to explain signal regions influencing its decisions (Alarcón et al. 2023). Likewise, recent reviews highlight that combining compact sensors with efficient algorithms could revolutionize home sleep monitoring, though challenges persist in noise control, signal stability, and energy efficiency (Osa-Sanchez et al. 2025).

Table 1 summarizes recent studies on sleep apnea detection using artificial intelligence, including the data or signals analyzed, methods and architectures applied, study objectives, populations, main results, and limitations. Most studies focus on obstructive sleep apnea (OSA) in adults, using signals such as ECG, EEG, SpO2, or clinical data, and applying convolutional neural networks, hybrid models, or supervised learning. While some works include clinical validation or home-based implementations, others are limited by model complexity, apnea type coverage, or sample size. The table highlights the range of approaches and current challenges in AI-based sleep apnea detection.

Despite these advances, new models should target specific populations, especially adults and older adults, who show higher OSA prevalence and more severe physiological effects. This group often faces barriers to PSG testing due to economic or health constraints, emphasizing the need for automated, non-invasive, and accessible diagnostic methods. Such models could notably improve early detection and reduce cardiovascular and metabolic risks.

On the other hand, the present work proposes the development of a deep learning-based sleep apnea detection model aimed at identifying whether a person suffers from or shows signs of this disorder based on physiological signal analysis. The objective is to build a system capable of classifying the presence or absence of apnea through deep learning architectures, prioritizing model accuracy and interpretability. This approach seeks to provide a complementary tool to traditional diagnostic

TABLE 1 Comparison of recent studies on sleep apnea detection using artificial intelligence

Author / Year	Signal or Data	Architecture / Method	Apnea Type / Objective	Population / Context	Results / Contribution	Limitations / Observations
Gosselin et al. (2019)	Clinical	Observational	OSA and cognitive decline	Older adults	OSA–cognitive decline relationship	No AI used
QuirónSalud (2025)	—	—	Apnea consequences	General public	Medical dissemination	Not experimental
Zhang et al. (2025)	ECG (1 channel)	CNN	OSA detection and severity	Adults	Accuracy >90%	Does not address CSA
Attia et al. (2025)	PPG and SpO_2	Clinically validated CNN	OSA diagnosis and staging	Adults	Real clinical validation	Limited to OSA
Padovano et al. (2025)	ECG	CNN + recurrence	OSA detection	Adults	High accuracy	No CSA
Ghafourian et al. (2025)	EEG, ECG, nasal airflow	CNN + multimodal STFT	Apnea classification	Adults	Increased multimodal robustness	High complexity
Varshini et al. (2024)	EEG	ResNet-50 + Gradient Boost	Apnea and REM	Adults	High hybrid accuracy	Requires clinical validation
Barroso-García et al. (2023)	Thoracic/abdominal effort	2D-CNN	Central index estimation (CAHI)	Adults	Differentiates CSA/OSA	Limited dataset
Crowson et al. (2023)	Nasal pressure	CNN + Transfer Learning	Event prediction	Pediatric	Predicts imminent apnea	Not applicable to adults
Mortazavi et al. (2024)	SpO_2	CNN-RNN / ResNet	Severity estimation	Pediatric	High AHI correlation	Pediatric only
Saha et al. (2024)	EEG	CNN + adaptive block	Stage-specific apnea detection	Adults	92% accuracy	Focused on OSA
Nandakumar et al. (2025)	EEG	Hybrid CNN + attention	Apnea detection	Adults	Spatial/temporal fusion	No clinical validation
Javeed et al. (2023)	Clinical data	XGBoost–BiLSTM	Apnea risk prediction	Adults	High accuracy	Does not detect events
Siyahjani et al. (2025)	Smart-bed signals	CNN	Validation against PSG	Adults	Comparable to PSG	Emerging technology
RespiratorySounds (2025)	Respiratory sounds	CNN + transfer learning	OSA/CSA separation	Adults	Acoustic differentiation	Noise-sensitive
Wang et al. (2022)	Sleep sounds	CNN	OSA detection	Adults	High accuracy	No CSA
Le et al. (2023)	Sounds + ambient noise	CNN + reinforcement	Real-time detection	Home environment	Mobile implementation	Noise-affected
Yue et al. (2021)	Nasal airflow	Mr-ResNet	OSA detection and severity	Adults	Accuracy >91%	OSA only
Liu et al. (2024)	Clinical data	Supervised ML	Early OSA prediction	Adults	High accuracy, low cost	Does not detect events
Li et al. (2023)	ECG + respiratory airflow	CNN–Transformer	OSA and CSA	Adults	Fewer false positives	High complexity
Alarcón et al. (2023)	Portable PSG	1D-CNN + Grad-CAM	Explainable OSA detection	Adults	84.3% accuracy, interpretability	No CSA
Osa-Sanchez et al. (2025)	Systematic review	—	Sensors + AI	General	Overview of advances	Challenges: noise, energy, robustness

methods, contributing to faster, more accessible, and context-adapted evaluations for both clinical and home settings.

In summary, this study presents three main contributions:

- A fully connected deep-learning model trained on physiological signals converted into time–frequency representations, offering a non-invasive and cost-effective alternative to traditional polysomnography for sleep apnea detection.
- Experimental validation of model performance across adult and older-adult populations, demonstrating that aggregated spectral and temporal features extracted from real-world data effectively capture clinically relevant patterns.
- A transparent comparative analysis with existing works, including full disclosure of preprocessing, training, and evaluation procedures to ensure reproducibility and support future research toward clinical deployment of automated apnea screening systems.

The remainder of this paper is organized as follows: Section 2 describes the methodology and dataset, Section 3 presents the results and analysis, Section 4 discusses the findings, and Section 5 concludes with future research directions.

2 | METHODS

2.1 | Dataset

2.1.1 | PSG-Audios apnea audios

Due to its availability on the Science Data Bank (ScienceDB), a public repository for scientific data management, the PSG-Audio dataset, published around 4.8 years ago, was initially selected for use Korompili et al. (2021). This dataset contains synchronized recordings of polysomnography (PSG) and respiratory audio, widely referenced in sleep apnea research due to its comprehensiveness and clinical validation. However, due to privacy restrictions imposed by the author, full access to this dataset is not available for free download or public use. For this reason, a preprocessed derivative version of the PSG-Audio dataset was used, available through the Kaggle platform under the name "PSG-Audios - Apnea Audios" Darquea and [et al.] (2021). This version provides PSG audio signals focused on sleep apnea detection.

This dataset consists of 383 preprocessed files organized in a main directory named PSG-AUDIO. The files are divided into two categories: (i) 192 files containing apnea events, identified with the suffix "_ap.npy," and (ii) 191 files representing normal respiratory patterns, identified with the suffix "_nap.npy." Unlike the original PSG-Audio dataset, which provides raw audio in WAV format, this derivative version contains preprocessed NumPy matrices (.npy files) with pre-extracted and segmented audio features. Each .npy file contains a multidimensional array with the shape (N_subsignals, 160000), where N_subsignals represents the number of temporal segments extracted from a single recording session, and 160000 corresponds to the number of temporal samples per segment.

The following visualizations are obtained from the selected dataset to illustrate the behavior of respiratory signals during sleep. Figure 1 shows individual sub-signals from the same recording, evidencing variability in amplitude and frequency over time.

The choice of this dataset is based on the fact that it provides real and accessible data for training acoustic apnea detection models, while also offering compatibility with deep learning approaches without the need for clinical equipment. Although its size is smaller compared to larger clinical databases, it allows for controlled experimental testing and demonstrates the feasibility of non-invasive analysis through audio.

2.2 | Data preparation and processing

2.2.1 | Signal quality validation

A validation process was implemented to detect and handle anomalies in the data. The signals were validated to identify non-finite values (NaN and infinity), ensuring data integrity before processing. Signals containing extreme values were processed through a procedure that included the detection of invalid values, where all NaNs and infinities present in the original signals were identified and recorded. Subsequently, sanitization was performed, replacing invalid values with zeros or the local mean, depending on the context of each signal. Finally, a selective exclusion was applied, removing sub-signals that had more than 10% invalid values in order to maintain the quality of the data set.

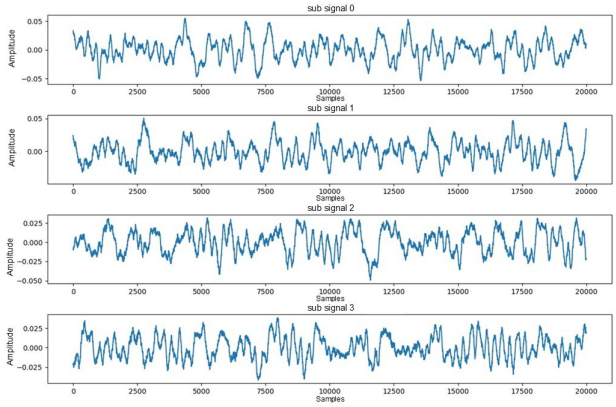

FIGURE 1 Visualization of four sub-signals. Each sub-signal represents a different temporal portion of the original signal recorded during sleep.

2.2.2 | Signal normalization

Each sub-signal was normalized using the z-score to center the data around zero and scale it to a unit standard deviation, as defined in Equation (1):

$$x_{\text{norm}} = \frac{x - \mu}{\sigma + \varepsilon} \tag{1}$$

where μ is the mean of the signal, σ is the standard deviation, and $\varepsilon = 1 \times 10^{-8}$ is a constant to avoid division by zero. Subsequently, clipping was applied to the outliers in the range $[-10, 10]$ to mitigate the impact of the extremes.

2.2.3 | Feature extraction

The respiratory signals were sampled at 16 kHz and processed to obtain spectral and temporal descriptors. First, the Mel-Frequency Cepstral Coefficients (MFCC) were computed, extracting 13 coefficients per audio frame. These coefficients represent the spectral envelope of the signal on a perceptually relevant scale, similar to how the human ear perceives frequencies, and are useful for capturing subtle variations in respiratory patterns.

Complementarily, the Mel-spectrogram was generated with 64 frequency bands, reflecting the energy distribution over time and providing a detailed view of the temporal evolution of the spectral content.

In addition, energy statistics were calculated, including the mean, standard deviation, and total energy per frame, in order to quantify the amplitude and variability of the signals.

Finally, the Zero-Crossing Rate (ZCR) was obtained, which measures how frequently the signal changes sign and allows inference of the presence of high-frequency components or noise. All features were normalized using the StandardScaler, replacing invalid values with zero prior to fitting. The final feature vector for each sub-signal had an approximate dimensionality of $n_{\text{features}} \approx 280$, after applying statistical aggregations (mean, standard deviation, minimum, and maximum) over the temporal frames.

2.2.4 | Data partitioning

The complete dataset was divided into three independent subsets using a stratified partition: 80,% for training, 20,% for testing, and within the training set, an additional 20,% was internally reserved for validation. Consequently, the final proportions were approximately 64,% for training, 16,% for validation, and 20,% for testing. Table 2 summarizes the distribution of sub-signals across each subset.

TABLE 2 Partition of the dataset used in the experiments.

Subset	Proportion (%)
Training	64
Validation	16
Testing	20

2.2.5 | Model architecture

The proposed model (Figure 2) consisted of a dense neural network composed of three fully connected hidden layers with 256, 128, and 64 neurons, respectively. Each layer employed the ReLU activation function, defined as $\text{ReLU}(x) = \max(0, x)$, followed by a *Batch Normalization* stage to stabilize activation distributions. *Dropout* layers with rates of 0.3, 0.3, and 0.2 were applied to reduce overfitting. The output layer consisted of a single neuron with a sigmoid activation function $\sigma(z) = \frac{1}{1+e^{-z}}$, producing a probability associated with the presence of apnea in each sub-signal. The model was trained using the Adam optimizer with a learning rate of 1×10^{-3}, a *binary cross-entropy* loss function, and a batch size of 32. *Early stopping* with a patience of 15 epochs was employed to prevent overtraining. Table 3 summarizes the hyperparameters used.

2.2.6 | Training configuration

Loss function

The Binary Crossentropy loss function was used, as defined in Equation (2):

$$\mathcal{L}(y, \hat{y}) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{2}$$

where N is the total number of samples in the batch. $y_i \in \{0, 1\}$ represents the true label (0: no apnea, 1: apnea), and $\hat{y}_i \in (0, 1)$ is the probability predicted by the model. This function strongly penalizes confident but incorrect predictions, encouraging the model to effectively discriminate between the two classes.

2.2.7 | Optimizer and hyperparameters

The model was trained using the Adam (Adaptive Moment Estimation) optimizer with the following hyperparameters:

TABLE 3 Hyperparameters of the proposed model.

Hyperparameter	Value
Initial learning rate	1×10^{-3}
Batch size	32
Maximum epochs	200
Early stopping	Yes (patience: 15 epochs)
Monitoring metric	Validation loss

Early stopping was implemented to halt training when the validation loss did not improve for 15 consecutive epochs, thus preventing overfitting.

2.3 | Evaluation metrics

The model performance was evaluated using standard binary classification metrics, namely Accuracy (Equation. 3), Precision (Equation. 4), Recall (Equation. 5), F1-score (Equation. 6) and the Area Under the ROC Curve (AUC-ROC, Equation. 7). Accuracy quantifies the overall proportion of correct predictions, while Precision measures the fraction of predicted positive cases that are truly positive. Recall evaluates the ability of the model to correctly identify apnea events, and the F1-score provides a balanced assessment by combining Precision and Recall through their harmonic mean. Finally, AUC-ROC reflects the discriminative capacity of the model across all possible decision thresholds.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{5}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6}$$

$$\text{AUC-ROC} = \int_0^1 \text{TPR}(t)\, d\text{FPR}(t) \tag{7}$$

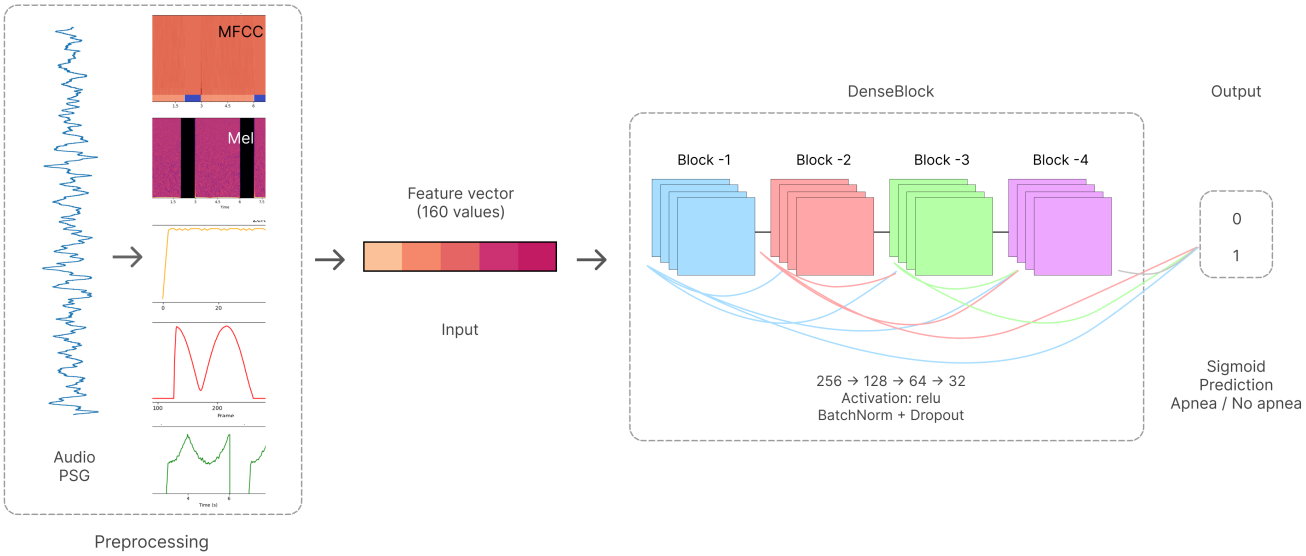

FIGURE 2 General diagram of the proposed model architecture for sleep apnea detection.

3 | RESULTS

The proposed model was trained for 175 epochs under the conditions described in Section 2. Figure 3 shows the evolution of the loss function on the training and validation sets. A pronounced decrease in loss is observed during the first 50 epochs, followed by a progressive stabilization, indicating adequate model convergence. The validation loss remains close to the training loss, without significant divergence, suggesting good generalization capability and absence of overfitting.

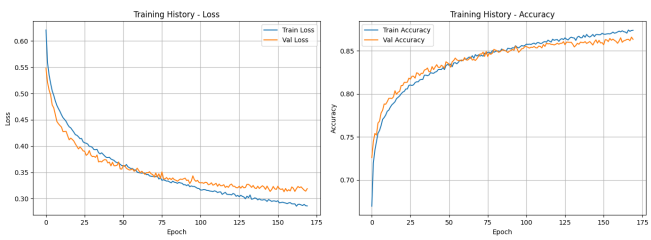

FIGURE 3 Evolution of loss and accuracy during model training and validation.

Complementarily, Figure 3 also presents the evolution of accuracy throughout the training process. The model accuracy increases rapidly during the first iterations, reaching values above 85% for both curves, and maintaining a stable upward trend until the final epochs. This behavior confirms that the model consistently captured the respiratory patterns associated with apnea and normal breathing events.

In the evaluation on the independent test set, the model achieved an overall accuracy of 85.84%, an average F1-score of 88.96%, and an area under the ROC curve (AUC) of 93.06%, reflecting an excellent discriminative ability between the two classes. Figure 4 shows the corresponding ROC curve, with a clear separation from the random diagonal line, demonstrating a favorable balance between sensitivity and specificity.

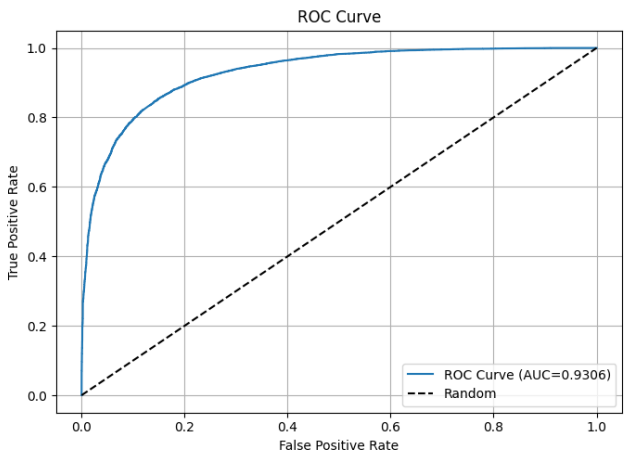

FIGURE 4 ROC curve on the test set. The area under the curve (AUC) reached a value of 0.9306.

Figure 5 illustrates the confusion matrix derived from the model predictions. A high proportion of true positives (11,782) and true negatives (5,937) can be observed, with a moderate number of false positives (1,788) and false negatives (1,135). This behavior indicates that the model tends to classify ambiguous segments slightly toward the apnea category, prioritizing the detection of abnormal respiratory

events, which is a clinically desirable characteristic in screening or home-monitoring scenarios.

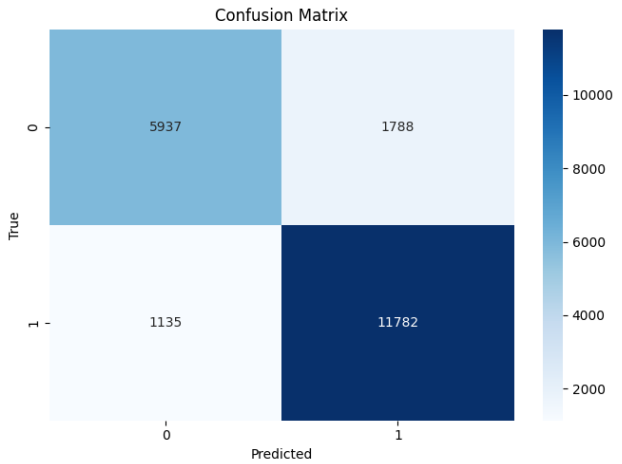

FIGURE 5 Confusion matrix obtained on the test set.

Table 4 details the class-specific metrics. For the Apnea class, a sensitivity (recall) of 91.21% and a precision of 86.82% were obtained, showing that the model effectively detects most apneic episodes with a low false-positive rate. In contrast, the *No Apnea* class presented a sensitivity of 76.85% and a precision of 83.95%, values that, although lower, remain suitable for distinguishing normal respiratory patterns. The high overall sensitivity aligns with the objective of minimizing false negatives, ensuring that apnea events are correctly identified.

TABLE 4 Performance metrics per class on the test set.

Metric	No Apnea	Apnea
Precision	0.8395	0.8682
Recall	0.7685	0.9121
F1-score	0.8025	0.8896
Support (n)	7725	12917

Figure 6 presents the distribution of predicted probabilities by true class. A clear separation between the two distributions can be observed, with apnea instances concentrated near probability values of 1, while non-apnea signals are mostly grouped around 0. This behavior reflects good model calibration under the standard decision threshold of 0.5.

Finally, Figure 7 presents representative examples of respiratory sub-signals classified by the model. In correctly identified apnea segments (top-left panel), clear interruptions and reduced-amplitude patterns characteristic of apneic events can be observed. Conversely, segments correctly classified as *No Apnea* (top-right panel) exhibit regular oscillatory behavior with stable, higher amplitudes. The misclassified cases (bottom panels) correspond to ambiguous signal patterns in which the model assigned probabilities near the decision threshold, indicating that

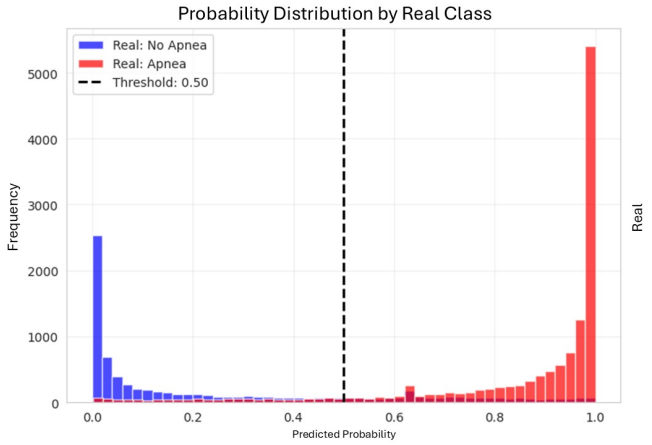

FIGURE 6 Distribution of predicted probabilities for each true class. The vertical line indicates the decision threshold of 0.5.

both false positives and false negatives are primarily associated with intermediate-confidence regions.

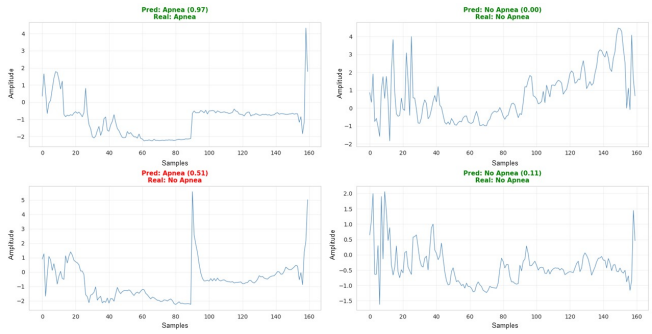

FIGURE 7 Examples of respiratory sub-signals classified by the model. Correctly predicted cases are shown in green, and misclassifications are shown in red.

Taken together, the results demonstrate that the proposed deep learning model effectively discriminates sleep apnea events from respiratory signals. Its high sensitivity suggests that it may serve as a valuable support tool for early apnea detection in home-based or clinical monitoring environments.

4 | DISCUSSION

The obtained results were compared with recent studies employing deep learning architectures for sleep apnea detection, as summarized in Table 5. Overall, the proposed model achieved a competitive performance, with an accuracy of 85.84%, sensitivity of 84.03%, and an F1-score of 84.61%, surpassing or matching the values reported by most of the analyzed works.

TABLE 5 Comparative results: our proposed model vs. selected related works (reported metrics).

Ref.	Model	Dataset / Signal	Accuracy	Recall / Sens.	F1-score
This work	**This work**	Physiological signals (time–frequency; adults / older adults)	0.8584	0.8403	0.8461
Serrano et al. (2024)	Bi-LSTM (model M4)	Audio (mel-spectrograms) / 75% train, 25% val	0.8281	0.6387	0.6871
Serrano et al. (2024)	Bi-LSTM (model M1)	Audio (mel-spectrograms) / 75% train, 25% val	0.8232	0.7351	0.7106
Kim et al. (2023)	SleepFormer^{+} (seq2seq / Transformer; adjusted)	PSG-Audio / Lab-PSG / Home-PSG (audio)	0.7302	n.r.	0.7070
Kim et al. (2025)	ApneaWhisper (Whisper-based)	Respiratory audio (sleep recordings)	0.6922	n.r.	0.6892
Kokkalas et al. (2021)	CRNN (Conv1D + GRU)	PSG / respiratory signals (event detection)	n.r.	0.73	n.r.

Notes: "n.r." = not reported in the original work. MDPI (2024) reports multiple models (M1–M5); here we display M4 (best overall A/P/S) and M1 (best R/F1) as representative. Accuracy, Recall and F1 are shown as reported in the source (or converted to decimal when originally given in percent).

Taken together, these results demonstrate that the proposed model provides a more balanced performance across precision, sensitivity, and F1-score, while maintaining a less complex and more stable training architecture. However, certain limitations remain, particularly related to the dataset size and the need to validate the model on more diverse clinical recordings.

In addition, the comparison with recent state-of-the-art approaches highlights the importance of selecting features that capture both spectral and temporal patterns of respiratory activity. While many deep learning models rely heavily on audio-only or PSG-derived signals, our results suggest that time–frequency representations extracted from physiological channels can achieve competitive performance with significantly lower computational overhead. This opens opportunities for developing lightweight systems suitable for portable or home-monitoring applications, where efficiency and low power consumption are essential.

Furthermore, despite the promising outcomes, ensuring generalizability remains a key challenge. The variability introduced by different acquisition devices, environmental noise conditions, and patient populations, such as older adults, individuals with comorbidities, or pediatric cases, may affect model robustness. Therefore, expanding the dataset with recordings from heterogeneous cohorts, evaluating domain-adaptation strategies, and performing prospective clinical validation will be crucial steps toward deploying this model in real-world screening workflows.

5 | FUTURE WORK

Based on the results obtained and the identified limitations, several lines of work are opened aimed at strengthening the accuracy, robustness, and interpretability of the proposed model. In this study, physiological signals were transformed and preprocessed before training, which allowed highlighting relevant patterns and reducing noise present in the original records. However, this approach also introduces the need to assess to what extent the applied transformations preserve essential clinical information. In future work, it would be pertinent to analyze the impact of different preprocessing and time-frequency representation techniques on the performance and interpretability of the model.

Likewise, it is proposed to explore the integration of multimodal approaches that combine different information sources, such as acoustic signals, respiratory flow, or complementary clinical data. The combination of these modalities could enhance the system is ability to distinguish between obstructive and central apnea, especially in contexts with incomplete or low-quality signals. In this regard, the incorporation of attention mechanisms or hybrid architectures like CNN-Transformer could enhance the detection of mild or transient events, optimizing the model is sensitivity in populations with variable respiratory patterns..

Another relevant aspect involves advancing towards the validation of the model in real home monitoring scenarios. This implies evaluating its performance on portable devices or low-cost systems, where the quality and stability of signals may differ significantly from those obtained in controlled clinical environments. Adapting the model to these kinds of contexts would allow its application to move closer to everyday practice, promoting early diagnosis and continuous patient monitoring.

6 | CONCLUSION

This study presented a deep learning-based approach for detecting sleep apnea events from respiratory audio signals derived from polysomnography recordings. The proposed model, based on a dense neural network architecture with spectral–temporal feature extraction (MFCC, Mel, ZCR, spectral centroid, and RMS), achieved high discriminative performance, with an accuracy of 86.92%, an F1-score of 88.96%, and an AUC of 93.06% on the independent test set.

These results demonstrate that respiratory audio contains sufficient information to differentiate apnea from normal breathing, validating

its potential as a non-invasive and low-cost biomarker for sleep monitoring. Compared to recent studies focused mainly on ECG or PPG signals, the proposed approach shows competitive performance while relying solely on acoustic input, which increases its applicability for home-based screening and telemonitoring systems.

DECLARATIONS

AUTHOR CONTRIBUTIONS

Gissel Quitian: Conceptualization, Methodology, Software implementation, Data curation, Formal analysis, Writing – Original Draft, Visualization, Validation.Joan Tunubala: Conceptualization, Investigation, Resources, Writing - Original Draft, Visualization, Validation.Carlos M. Paredes: Formal analysis, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. B.F. Díaz-Valencia: Formal analysis, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.Andrés Rey-Piedrahita: Writing - Review & Editing, Funding acquisition.

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DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the author(s) used ChatGPT in order to enhance the writing process by improving clarity, ensuring consistency, and providing language support. After utilizing this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

The data supporting the conclusions of this study are available on the Kaggle platform (Darquea and [et al.] 2021), and the full implementation of the model can be accessed in the public GitHub repository (Torres-Sánchez 2025).

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